

A Stronger Rigidity Formulation for the Zaks Cycle

Main point

The current theorem about the Zaks Hamiltonian cycle can be reframed as a sharp case of a more general rigidity principle for cyclic orderings of S_n .

The general principle does not depend on the pancake graph. It only uses the left regular action of S_n and the elementary fact that the centralizer of an n -cycle in S_n is the cyclic group generated by it.

General cyclic-order rigidity theorem

Let C be a cyclic ordering of the elements of S_n . Equivalently, C is a Hamiltonian cycle on the complete graph with vertex set S_n ; no Cayley graph structure is needed here.

For $g \in S_n$, write L_g for left multiplication:

$$L_g(\sigma) = g\sigma.$$

Theorem 1 (Cyclic-order rigidity for S_n). *Let $n \geq 3$. Let C be a cyclic ordering of the elements of S_n , and let $c \in S_n$ be an n -cycle. Assume that C is invariant under the left translation L_c :*

$$L_c(C) = C.$$

Then

$$\text{Stab}_{L(S_n)}(C) \subseteq \{L_g : gcg^{-1} \in \{c, c^{-1}\}\}.$$

In particular,

$$|\text{Stab}_{L(S_n)}(C)| \leq 2n.$$

Proof. Since C is a cyclic ordering, its abstract automorphism group is dihedral. The translation L_c acts on C as a rotation of order n . If L_g stabilizes C , then inside the automorphism group of the cycle it conjugates this rotation either to itself or to its inverse:

$$L_g L_c L_g^{-1} = L_c \quad \text{or} \quad L_g L_c L_g^{-1} = L_c^{-1}.$$

The left regular action is faithful, hence

$$gcg^{-1} = c \quad \text{or} \quad gcg^{-1} = c^{-1}.$$

It remains to count the possible g 's. The centralizer of an n -cycle in S_n is exactly

$$C_{S_n}(c) = \langle c \rangle.$$

Thus there are n elements with $gcg^{-1} = c$. The elements with $gcg^{-1} = c^{-1}$ form one coset of this centralizer, hence there are another n of them. Therefore the stabilizer is contained in a set of size $2n$. $\square$

The containing group is the dihedral group of the n -gon whose cyclic order is defined by c . If

$$c = (a_1 a_2 \cdots a_n),$$

then it is the group of symmetries of the polygon

$$a_1, a_2, \dots, a_n.$$

Sharpness: the Zaks cycle attains the bound

Now return to the pancake graph

$$P_n = \text{Cay}(S_n, \{r_2, \dots, r_n\}),$$

with generators acting by right multiplication. Let Z_n be the Zaks Hamiltonian cycle, and set

$$\rho = r_{n-1}r_n.$$

The Zaks recursion gives

$$L_\rho(Z_n) = Z_n.$$

Thus the general theorem applies with $c = \rho$.

Moreover, the palindromic property of the Zaks word gives the reflection

$$L_{r_n}(Z_n) = Z_n.$$

Since

$$r_n \rho r_n = \rho^{-1},$$

we get the full dihedral subgroup

$$\langle L_\rho, L_{r_n} \rangle \cong D_n$$

inside the left-regular stabilizer.

The general rigidity theorem gives the reverse inclusion, because no left-regular stabilizer of a cyclic ordering invariant under L_ρ can have more than $2n$ elements. Therefore

$$\text{Stab}_{L(S_n)}(Z_n) = \langle L_\rho, L_{r_n} \rangle \cong D_n.$$

Corollary 1 (Sharpness). *The bound*

$$|\text{Stab}_{L(S_n)}(C)| \leq 2n$$

is sharp. It is attained by the Zaks Hamiltonian cycle in the pancake graph.

Minimal Coxeter interpretation

This is the Coxeter/Weyl dictionary, in only the amount needed for the paper.

The group S_n is the Weyl group of type A_{n-1} . Its simple Coxeter generators are the adjacent transpositions

$$s_i = (i \ i+1).$$

The pancake generator r_k is not a simple reflection. Instead,

$$r_k = \text{the longest element of the parabolic subgroup } \langle s_1, \dots, s_{k-1} \rangle \cong S_k.$$

In particular,

$$r_n = w_0,$$

the longest element of the Weyl group S_n .

Also,

$$\rho = r_{n-1}r_n$$

is an n -cycle, hence a Coxeter element of type A_{n-1} . Thus the stabilizer found above is not an arbitrary dihedral group:

$$\langle \rho, r_n \rangle = \langle c, w_0 \rangle,$$

the natural dihedral group generated by a Coxeter element and the longest element.

Remark 1. *For readers not interested in Coxeter theory, the whole argument can be read purely in terms of S_n : n -cycles give the natural rotations, and the centralizer of an n -cycle is the cyclic group it generates. The Coxeter language explains why the Zaks choices $\rho = r_{n-1}r_n$ and r_n are geometrically distinguished.*

Effect on the paper

This reframing changes the structure of the proof.

1. Prove the general cyclic-order rigidity theorem for S_n .
2. Prove that the Zaks recursion gives $L_\rho(Z_n) = Z_n$.
3. Prove that the palindrome gives $L_{r_n}(Z_n) = Z_n$.
4. Conclude exactness from the general rigidity theorem.
5. Use Deng–Zhang to pass from the left-regular stabilizer to the full graph automorphism stabilizer for $n \geq 5$.

So the Zaks theorem becomes: a general upper bound for cyclic orderings of S_n , plus a natural pancake-graph construction that attains the bound.